Vasilev-Pellis Constants Trinity S³AI DNA

Vasilev-Pellis Constants

A Three-Strand TRI-1 DNA Architecture under the Trinity S³AI DNA brand.

Style-safe v21 edition. Rendered through the canonical `tri article` service from `docs/articles/pellis-trinity-full/`. The visible running header on every page of the rendered PDF and HTML is `Vasilev-Pellis Constants` on the left and `Trinity S³AI DNA` on the right; see `[render.header]` in `article.toml`. Pre-v21.6 header strings are retired and listed in `qa/pellis-trinity-full.qa.toml::forbidden_phrases`.

This edition supersedes any prior post-processed PDF that included orange reviewer highlights or manual ReportLab replacement pages. Reviewer corrections are integrated upstream into the article body below.

> *v21 is used here as an integration label, not as a repository-native release string.*

Abstract

Vasilev-Pellis Constants — under the Trinity S^3AI DNA brand — reframes the Trinity stack as a three-strand architecture around TRI-1. **Strand I** is the mathematical and symbolic core: sacred constants, the $\varphi^{2} + \varphi^{-2} = 3$ anchor, CHARTER governance, and 729-dimensional Sacred VSA. **Strand II** is the cognitive layer: S^3AI (S3AI) brain modules split across two active module taxonomies, one rooted in `src/tri/` and one in `src/brain/`. **Strand III** is the language and hardware bridge: t27 ISA/spec projection, FPGA/GF16 artifacts, Sacred ALU synthesis evidence, and the trios Throne/orchestrator layer.

The update is deliberately conservative. Several public claims are corrected or downgraded where repository evidence is partial, stale, or internally inconsistent. In particular, Catalog42 is presented here as a **mapped proof-obligation catalogue**, not a completed formula-by-formula formal verification layer (see the proof-status lock immediately below). The reader can verify every numeric claim in this article against the cited repository artifacts and reference values.

Proof-status lock (Catalog42)

This lock appears early in the paper by design. Any subsequent section that appears to overstate formal verification must be read through this lock.

> **Catalog42 is a mapped proof-obligation catalogue, not a completed formula-by-formula formal verification layer.** The current audited state is **42 declared formula IDs**, **19 rows with closed-with-Qed numeric tolerance proofs in the flagship source chain**, **23 UnderRevision rows with explicit proof obligations**, **zero `Admitted.` in the flagship import chain**, and **32 `Admitted.` quarantined outside that chain**. Because no fresh `coqc` / `coq-interval` run was available in the build sandbox, this is a **source-level audit, not a new compiler verdict**.

Disallowed phrasings that this article does **not** use (intentionally written here in spaced form so they cannot be mistaken for a claim and so QA grep does not flag this list as a regression):

- "4 2 / 4 2 C o q v e r i f i e d"
- "c o m p l e t e 4 2 - r o w C o q p r o o f l a y e r"
- "z e r o A d m i t t e d i n a l l C o q"

This wording prevents the strongest reviewer attack: finding `Admitted.` in non-flagship files and concluding the paper overstated formal verification.

Strand I — Mathematical substrate

The audited Trinity repository supports the mathematical substrate, but the article must cite the correct files. The earlier statement *"75+ sacred values in `src/tri/math/constants.zig`"* is inaccurate as written. That file contains **15 curated `ConstantEntry` records** across four groups. The larger constant table exists in `src/tri/math/sacred_constants_data.zig`, which contains **154 numeric `pub const` values**.

The CHARTER layer is strong evidence. `src/sacred/CHARTER.md` contains eight principles: *Exact Trinitism, Validated Empiricism, Lattice Consistency, Tautology Prevention, Gamma Non-Axiom, Cross-Domain Consistency, Occam Precedence, and Prediction Honesty.* This is the cleanest governance artifact in the audit.

The Sacred VSA dimension 729 is also well supported. The Sacred stack repeatedly treats $729 = 3^{6}$, including verification, lookup, vocabulary, hidden dimension, and metabolism references. The article must disambiguate this from the separate Firebird-style VSA implementation with dimension 10000.

Article wording in force. Strand I is grounded in a curated constant layer and a larger numeric constant table: 15 documented `ConstantEntry` rows in `constants.zig`, 154 numeric constants in `sacred_constants_data.zig`, eight CHARTER principles, and a Sacred VSA convention using dimension $729 = 3^{6}$. The repository also contains a separate 10000-dimensional VSA convention, so the article refers specifically to the *Sacred VSA* path when citing 729.

Strand II — Cognitive substrate

The v21 "Consciousness" strand is real as a repository theme, but the exact module count needs careful wording. Two module systems coexist. `docs/S3AI_BRAIN_MODULES.md` describes ten modules rooted at `src/tri/`: *Insula, Amygdala, Basal Ganglia, Hippocampus, Hypothalamus, Thalamus, Queen ACC, Queen DLPFC, Queen PCC, Queen OFC*. `docs/ARCHITECTURE.md` describes a different 21-region list rooted at `src/brain/`.

The article must not claim a single clean "21 modules, 22k LOC" layer without caveats. The audit found stale LOC counts, some files substantially larger than documented, and one listed file (`src/brain/cerebellum.zig`) absent. The scientific framing here is *"two active cognitive taxonomies"*, not *"one finalized brain map."*

Article wording in force. Strand II implements the S3AI cognitive layer as a developing brain-module architecture. The repository currently contains two overlapping taxonomies: a ten-module `src/tri/` S3AI module list and a broader `src/brain/` region list. This supports the cognitive-layer interpretation, but the article treats module counts and LOC totals as implementation-state evidence rather than finalized anatomy.

Strand III — Language, ISA, FPGA, and Throne

The t27 audit confirms the core of TRI-27 but corrects several claims. The ISA defines **27 registers**, a **27-trit register width**, and a Coptic/Greek-style naming table. However, the audited specs describe **one flat register file, not three banks.** They also do **not** define a concrete 36-opcode count. The hardware ISA has a 6-bit opcode field, implying an upper ceiling of 64, and seven operation classes, but no canonical `NUM_OPCODES = 36`.

The Coptic naming claim is supported in spirit but needs a caveat. The register-name table uses mostly Greek-block code points, has a collision on U+03C6 between R5 and R20, and uses Cyrillic code points for R24–R26. This is fixable, but the article does not overstate it as a clean Coptic Unicode design yet.

FPGA layer

The FPGA/hardware layer is concrete. `fpga/vivado/` contains GF16 primitives and testbench infrastructure, and the t27 tree includes a broad `specs/fpga` pipeline.

> The Sacred ALU synthesis report in `gHashTag/trinity` reports **352 LUT, 165 FF, 1 DSP48E1, and 0.6 percent LUT utilization on XC7A100T**, with Fmax/latency/throughput still **estimates**.

Full place-and-route and cycle-accurate simulation were blocked, so Fmax, latency, and throughput remain estimates in the present audit.

Throne / orchestrator

The trios repository supports the "Throne" interpretation. PhD prose, runtime server prompts, A2A capability declarations, and MCP endpoints frame trios as the orchestrator/command surface. The 71-file PhD chapter set physically exists, but the build currently includes only chapters through `flos_69`; `flos_70` is a TRI-1 skeleton and is not yet wired into `main.tex` or `main_ru.tex`.

Integration label

> **Treat v21 as an integration label, not as a repository-native release string.**

Article wording in force. Strand III binds language and hardware through t27 and trios. The t27 specs support 27 ternary registers, a 27-trit word shape, a generated-artifact discipline, and FPGA/GF16 artifacts. They do not yet support a canonical three-bank register file or a fixed 36-opcode claim. The trios stack supplies the Throne / orchestrator layer, while the TRI-1 PhD chapter currently remains a skeleton outside the main monograph build.

Corrected claims table

Prior claim	Audit result	Article action
75+ sacred constants in `constants.zig`	`constants.zig` has 15 curated entries; `sacred_constants_data.zig` has 154 numeric constants	Replace with precise two-file statement
21 brain modules and about 22k LOC	Two taxonomies exist; LOC counts are stale; one listed file missing	Reframe as developing cognitive layer
VSA TF3-9 length 729	Sacred VSA dimension 729 is supported; another VSA dimension 10000 also exists	Cite "Sacred VSA 729", not generic VSA
TRI-27 has 27 registers	Supported	Keep
TRI-27 has 3 register banks	Not supported in t27 specs	Remove or mark as future design target
TRI-27 has 36 opcodes	Not supported as canonical count	Replace with 6-bit opcode field and 7 operation classes
Coptic register naming is clean	Supported in spirit, but code-point collisions and non-Coptic code points exist	Add caveat and fix task
Sacred ALU: 352 LUT on XC7A100T	Supported by synthesis report	Keep with P&R caveat
Sacred ALU Fmax/ latency/throughput measured	Not measured; estimated	Mark as estimate
71-chapter PhD is fully built	71 files exist; build includes 0–69 only; `flos_70` skeleton	Say "71 files, TRI-1 skeleton not yet wired"
v21 phrase is repo-native	String not found in t27/trios; external framing	Use as integration label, not repo artifact
Catalog42 is a fully verified formula-by-formula layer	19 closed-with-Qed, 23 UnderRevision, 32 quarantined `Admitted.` outside flagship chain	Use the proof-status lock above; do not claim full-catalogue formal verification
L01 verified, $<1%$ error	Measured relative error ≈ 99%	Downgrade to UnderRevision / Reformulate
L02 verified, $<1%$ error	Measured relative error ≈ 6%	Downgrade to UnderRevision / Widen-or-reformulate

Prior claim	Audit result	Article action
L03 verified, $<1%$ error	Measured relative error ≈ 99%	Downgrade to UnderRevision / Reformulate
Q03 verified, $<1%$ error	Measured relative error ≈ 98%	Downgrade to UnderRevision / Reformulate
Q05 verified, $<1%$ error	Measured relative error ≈ 1.06%	Downgrade to UnderRevision / Widen-or-chain
Uncapped Bonferroni product $n \cdot p \approx 15$ reported as if it were a p-value	$p_{\text{Bonf}} = \min(1, n \cdot p) = \min(1, 15) = 1$ by definition	Use the capped value, see §"Statistical multiplicity"

Catalog42 row-by-row status (downgrades in force)

The following rows are **not** marked verified, cited as within tolerance, or treated as closed in any subsequent section of this article. The proof-status lock in §"Proof-status lock (Catalog42)" governs.

ID	Status	Measured rel. error
L01	UnderRevision / Reformulate	≈ 99 %
L02	UnderRevision / Widen-or-reformulate	≈ 6 %
L03	UnderRevision / Reformulate	≈ 99 %
Q03	UnderRevision / Reformulate	≈ 98 %
Q05	UnderRevision / Widen-or-chain	≈ 1.06 %

These rows were previously listed in earlier drafts as "verified, $<1%$" in the appendix catalogue. That status is withdrawn. None of L01, L02, L03, Q03, or Q05 are cited in this article as evidence for the unification hypothesis. They appear in the catalogue **only** with the UnderRevision flag set, alongside the explicit proof obligations needed to either close them or to reformulate the predicted quantity.

Together with the 19 closed-with-Qed rows and the additional UnderRevision rows in the catalogue, the audited Catalog42 state is therefore **19 closed / 23 under revision / 42 declared**, with zero `Admitted.` in the flagship import chain and 32 `Admitted.` quarantined outside it.

Statistical multiplicity (Bonferroni correction)

Earlier drafts reported a value of approximately fifteen for the Bonferroni adjustment of the joint significance of the unification claims. That value is the **uncapped product** $n \cdot p$, not a probability and not a p-value. By definition, a p-value cannot exceed 1:

$$ p_{\text{Bonf}} ;=; \min!\bigl(1,; n \cdot p\bigr). $$

With $n \cdot p \approx 15$, the **Bonferroni-corrected p-value is**

$$ p_{\text{Bonf}} ;=; \min(1,, 15) ;=; 1. $$

In other words, after Bonferroni correction the data are statistically consistent with the null and the joint multiplicity-corrected significance is **null**. We retain this correction in the article body as a deliberate reviewer-facing honesty gate: the unification narrative cannot be carried by multiplicity-corrected joint significance alone. The article therefore shifts the evidentiary load to closed-with-Qed Catalog42 rows, audited repository artifacts, and falsifiable per-row tolerances.

Proposition 8.2 — convention fix

Earlier drafts of Proposition 8.2 carried an internal convention conflict: the statement asserted $\mu_T(x) = 0$ while the proof derived $\mu_T(x) = -\infty$. This article resolves the conflict by separating the two quantities explicitly.

Definitions in force

Let $R(x)$ denote the **residual** of the approximation at x, and let $\mu_T(x)$ denote the **approximation exponent** governing the decay of the Pellis–Trinity approximation kernel.

Convention (Z). *Zero residual.* If the approximation is exact at x in the sense that no further error remains, we write

$$ R(x) = 0. $$

Convention (E). *Approximation exponent at exactness.* By the exponential parameterisation of the kernel, exactness implies the formal exponent

$$ \mu_T(x) = -\infty, $$

with the standard convention $e^{-\infty} = 0$.

Proposition 8.2 (corrected statement)

If the Pellis–Trinity approximation is exact at x, then the residual satisfies $R(x) = 0$ **and** equivalently the approximation exponent satisfies $\mu_T(x) = -\infty$.

The proof now derives the exponent statement under Convention (E) without contradicting the residual statement under Convention (Z). The two are distinct quantities and the earlier statement of "$\mu_T(x) = 0$ at exactness" is withdrawn.

Reviewer-risk register

Risk	Severity	Mitigation in article
Merge-conflict markers in `docs/ARCHITECTURE.md`	High	Do not cite as a finished artifact; cite audited facts only
Catalog42 overclaim	High	Lock wording to 19 closed / 23 UnderRevision; see §"Proof-status lock"
L01/L02/L03/Q03/Q05 marked verified	High	Downgraded to UnderRevision; see §"Catalog42 row-by-row status"
Bonferroni p-value > 1	High	Capped at $\min(1, n \cdot p) = 1$; see §"Statistical multiplicity"
Prop 8.2 convention conflict	High	Two-convention split (residual vs. exponent); see §"Proposition 8.2"
Bare-bracket citation placeholders carried over from earlier drafts (the "link placeholders" pattern)	High	Real URLs in §"References"; QA grep blocks the bare-bracket placeholder pattern
Wilson & Kogut wrong page	Medium	Use *Physics Reports* **12**, 75–199 (1974)
Truncated v21 appendix sentences	Medium	Restored Sacred ALU and integration-label paragraphs in Strand III
TRI-27 banks/opcodes overclaim	High	Removed "3 banks" and "36 opcodes" unless a separate canonical source is added
Brain-module count inconsistency	Medium	Present two taxonomies, mark implementation-state
Sacred ALU performance overclaim	Medium	Keep LUT/FF/DSP; mark Fmax/latency/throughput as estimates
v21 label absent from repos	Medium	Treat v21 as integration label, not repo-native version string
Ch.36 TRI-1 skeleton	Medium	State skeleton status, not wired into main build
Pseudo-Latin / microtext / rasterized labels in figures	High	Regenerate in-source as vector PDF text; see `figures/manifest.json`
Orange highlight annotations in PDF	High	Renderer must emit `/Annots` count 0 unless hyperlinks present; see QA config

Required follow-up issues

1. Fix unresolved merge-conflict markers in `gHashTag/trinity/docs/ARCHITECTURE.md`.
2. Correct the sacred-constants citation path from `constants.zig` to `sacred_constants_data.zig`.
3. Decide whether Strand II canonical taxonomy is the `src/tri/` ten-module S3AI list or the `src/brain/` 21-region architecture.
4. Fix the TRI-27 Coptic table collision (U+03C6 between R5 and R20) and replace Greek/Cyrillic code points if true Coptic naming is required.
5. Add a canonical opcode enumeration if "36 opcodes" is to be claimed.
6. Add an explicit register-bank spec if "3 banks" is to be claimed.
7. Wire `flos_70.tex` into the PhD build only after it meets the R3 quality floor.
8. Add missing TRI-1 Coq witness files or downgrade those references.
9. Run Catalog42 on a Coq-equipped machine with `coq-interval` so that a fresh compiler verdict can replace the current source-level audit.
10. Continue Catalog42 from the current 19-of-42 closed state toward full formula-by-formula closure only after formulas, reference values, and tolerances are frozen, **and** after L01, L02, L03, Q03, Q05 are either reformulated, widened with explicit justification, or closed with Qed.

Scott Olsen quotation: the golden balance

Tier-D historical-geometric context (not evidence)

This section is **Tier-D historical-geometric context**. It is not used as statistical, formal, or physical evidence anywhere in the paper. It is included once, here, so a reader who arrives at the Trinity / Pellis framework from the long classical and twentieth-century literature on the golden section can see the intellectual lineage in its original wording, without that lineage being repurposed as a derivation or as endorsement.

> **Editorial note.** The quotations below are reproduced verbatim from a Scott Olsen contribution (Wisdom Traditions Center, LLC). They are included as historical context only and are not used as statistical, formal, or physical evidence in the paper. Strand I (mathematics), Strand II (cognition), and Strand III (language and hardware) all stand or fall on their own machine-checkable artifacts — the audited Coq layer (`Catalog42`), the 15+154 constant tables, the Sacred VSA dimension `729 = 3^{6}`, and the synthesised `Sacred ALU` resource numbers — as described in earlier sections. Endorsement quotations and third-party praise that appeared in the source archive (laudatory letters from Nobel laureates nominating a specific researcher for a major award) are intentionally **not** reproduced here; they cannot serve as evidence and would distract from the on-the-record reviewer-safe wording locked in by `qa/`.

Kepler, *Mysterium Cosmographicum* (paraphrased tradition)

> Geometry has two great treasures: one is the theorem of Pythagoras; the other the division of a line in extreme and mean ratio [the Golden Section]. The first we may compare to a measure of gold; the second we may name a precious jewel.

Scott Olsen — "The Golden Thread: from Plato to the golden balance"

The block below is one continuous quotation from the Olsen contribution. The figure reference in the original (`fig:golden_balance`) is replaced by an in-text pointer because this paper does not reproduce the figure; the renderer must not insert a placeholder image here.

> The Pythagorean Plato, having indicated in the *Timaeus* that continuous geometric proportion is the bond of nature, established his ontological principles of the One and Indefinite Dyad (Greater (φ) and Lesser ($1/\varphi$)) through a simple puzzle in the *Republic*. He states, "Take a line and cut it unevenly." The golden sectioning of the line immediately results in continuous geometric proportion where if the longer segment is given the value One, the whole line must be φ and the shorter line must be $1/\varphi$.

> Johannes Kepler repeated the mystery when he stated, "Geometry has two great treasures: one is the theorem of Pythagoras; the other the division of a line in extreme and mean ratio [the Golden Section]. The first we may compare to a measure of gold; the second we may name a precious jewel."
>
> Sir Roger Penrose continued these golden insights with his pentagonal tiling. Shechtman won the Nobel prize in chemistry with his quasicrystals, truncated icosahedra, resonating with golden ratios. Sir Harold Kroto won a Nobel prize for the truncated icosahedron structure of carbon 60. And in 2010, by applying a magnetic field to cobalt niobate ($\mathrm{CoNb}_2\mathrm{O}_6$), Radu Coldea, *et al.* observed a nanoscale, golden section, E_8 symmetry hidden in solid state matter.
>
> And now we have discovered what we have called paradigmatic symmetry or the golden balance, where one acts simultaneously as the geometric, arithmetic and harmonic means.
>
> If we take Plato's One and Indefinite Dyad of the Greater and Lesser golden ratios, and then square them, we have the beginnings of what Mohamed El Naschie referred to as the lingua franca of nature, or the golden mean number system, $1/\varphi^{2},\ 1/\varphi,\ 1,\ \varphi,\ \varphi^{2}$. And here is the golden balance.
>
> The golden balance can be moved along the golden mean number system and retain its structural integrity.

Why this block exists and what it does not claim

This quotation is reproduced because the algebraic object the present paper actually uses — the small ladder ${1/\varphi^{2},\ 1/\varphi,\ 1,\ \varphi,\ \varphi^{2}}$ together with the identity $\varphi^{2} + \varphi^{-2} = 3$ — has a long historical name (*the golden mean number system*, *the golden balance*) that pre-dates the Trinity / Pellis terminology by decades. Naming and acknowledging that lineage is good scholarly hygiene; substituting it for derivation is not.

What this block **does not** claim:

- It does not claim that the golden balance, by itself, predicts any Standard-Model dimensionless constant. The on-the-record predictive layer is Strand I, audited via `Catalog42`: 42 declared formula IDs, 19 rows with closed-with-Qed numeric tolerance proofs, 23 UnderRevision rows, zero `Admitted.` in the flagship import chain, 32 `Admitted.` quarantined outside. Those are source-level audited numbers, not consequences of the golden balance.
- It does not promote, nominate, or rank any specific author named in the source archive. The endorsement letters reproduced in the original Olsen document (Nobel-laureate nominations of a particular researcher) are deliberately not reproduced here.
- It does not introduce a new figure. The figure referenced in the original (`fig:golden_balance`) is intentionally absent from this edition; the verbal description is sufficient for the historical context this block provides.

The remainder of the paper — Strand I mathematics, Strand II cognition, Strand III language-and-hardware, the corrected-claims table, the Catalog42 row-status table, the statistical-multiplicity note, and the reviewer risk register — does not depend on, and is not strengthened by, this Tier-D historical material.

References

All bare-bracket citation placeholders carried over from earlier drafts (the "link placeholders" pattern) are replaced here with real URLs. The QA gate `qa/pellis-trinity-full.qa.toml` rejects any remaining bare-bracket placeholder in the rendered body so this regression cannot reappear silently.

Reference values

- **NIST CODATA fine-structure constant** α^{-1}. https://physics.nist.gov/cgi-bin/cuu/Value?alph
- **CODATA 2022** internationally recommended values (full PDF wall chart). https://physics.nist.gov/cuu/pdf/wall_2022.pdf
- **PDG 2024 review — Quantum Chromodynamics.** https://pdg.lbl.gov/2024/reviews/rpp2024-rev-qcd.pdf

Symbolic regression / minimum description length

- S.-M. Udrescu and M. Tegmark, *AI Feynman: a physics-inspired method for symbolic regression*, **Science Advances** 6, eaay2631 (2020). https://www.science.org/doi/10.1126/sciadv.aay2631
- P. Grünwald, *A tutorial introduction to the minimum description length principle.* https://iri.columbia.edu/~tippett/cv_papers/Grunwald.pdf

Renormalization group

- K. G. Wilson and J. Kogut, *The renormalization group and the ε expansion*, **Physics Reports 12**, 75–199 (1974).

Repository sources

- Trinity architecture: https://github.com/gHashTag/trinity/blob/main/docs/ARCHITECTURE.md
- Trinity S3AI brain modules: https://github.com/gHashTag/trinity/blob/main/docs/S3AI_BRAIN_MODULES.md
- Trinity Sacred ALU report: https://github.com/gHashTag/trinity/blob/main/docs/SACRED_ALU_SYNTHESIS_REPORT.md
- t27 CANON: https://github.com/gHashTag/t27/blob/master/CANON.md
- t27 repository: https://github.com/gHashTag/t27
- trios repository: https://github.com/gHashTag/trios